✕ Catching a bug in a block costs more than it does in a system * 0/1

- (●) True ✕
- ( ) False

Correct answer

- (●) False

✓ Number of spins for a design is * 1/1

- ( ) The number of edits on a design from the customer
- (●) The number of times the chip went through the tape-out process ✓
- ( ) None of the above

✓ The most efficient way for a project is * 1/1

- ( ) Start with the design then verification
- ( ) Start with design and verification at the same time
- (●) Start with planning for verification firstly then going through design and verification in parallel ✓

✓ The tasks in System Verilog are always synthesisable * 1/1

- ○ True
- ◉ False ✓

✓ A non-driven output of type bit gives X on the simulation * 1/1

- ○ True
- ◉ False ✓

✓ The logic data type is a 4-state type and unsigned by default * 1/1

- ◉ True ✓
- ○ False

✓ The code coverage tests how many times you accessed each statement in the design through your test bench *1/1

- ◉ True ✓
- ○ False

✓ The code coverage tests if your design is functioning right * 1/1

- ○ True
- ◉ False ✓

✓ A do file is * 1/1

- ◉ Automating script for the tool ✓
- ○ Replacement for the testbench
- ○ Encrypted database

✓ We ignore the code coverage details of the ... in the code coverage report *1/1

- ○ Design
- ◉ Testbench ✓

✓ The default return data type in function is * 1/1

- ○ Int
- ○ Reg
- ◉ Logic ✓

✓ typing " vcover report *****.ucdb -details -annotate -all -output ******.txt " *1/1

- ○ Before simulation
- ○ After running simulation
- ◉ After ending simulation ✓

✗ It is recommended to declare the signal that will be connected to the DUT output port in the testbench to be of .... data type *0/1

- ○ 2-state
- ○ 4-state
- ◉ Any of them ✗

Correct answer

- ◉ 4-state

✕ The main root cause of ASIC functional flaws * 0/1

- ○ Design Error
- ○ Changing specification
- ◉ Flaw in internal reused block cell magacell or ip ✕

Correct answer

- ◉ Design Error

✕ Most of verification engineers spend time on * 0/1

- ◉ Test planning ✕
- ○ Testbench deverlopment
- ○ Debugging

Correct answer

- ◉ Debugging

✕ Dynamic verification doesn't use testbench * 0/1

- ◉ True ✕
- ○ False

✓ Code coverage measures the correctness of the design, not how thoroughly your tests exercised the design. *1/1

- ○ True
- ◉ False ✓

✓ In Branch coverage details, if/else coverage is reported * 1/1

- ◉ True ✓
- ○ False

✗ What is the acceptable percntage of coverage for critical applications *0/1

- ◉ 80 - 100 % ✗
- ○ 60 - 100 %
- ○ 100 %
- ○ none above

Correct answer

- ◉ 100 %

✓ Number of Bins for toggle coverage are.....(where n is number of bits in the DUT) *1/1

- (●) 2n ✓
- ( ) n
- ( ) n^2

✓ What is the primary reason for the increasing need for digital verification in IC design? *1/1

- (●) IC complexity is increasing ✓
- ( ) Transistor sizes are getting larger
- ( ) Verification is easier than design
- ( ) Manufacturing costs are decreasing

✓ Which language is predominantly used in modern IC verification? * 1/1

- ( ) Verilog
- ( ) VHDL
- (●) SystemVerilog ✓
- ( ) Assembly

✓ What is the primary goal of verification? * 1/1

- ○ Improve power efficiency
- ○ Reduce the number of transistors
- ◉ Detect design bugs before tape-out ✓
- ○ Increase clock frequency

✓ Which of the following is a 2-state data type in SystemVerilog? * 1/1

- ○ reg
- ○ logic
- ◉ bit ✓
- ○ wire

✓ What are the possible values of a 2-state bit in SystemVerilog? * 1/1

- ○ 0, 1, X, Z
- ◉ 0, 1 ✓
- ○ X, Z
- ○ 0, 1, Z

✓ What is the advantage of 2-state data types over 4-state data types * 1/1

- ○ It supports more states
- ◉ It uses less memory ✓
- ○ It allows X and Z values
- ○ It is required for combinational logic

✓ Which of the following SystemVerilog types is an upgrade to reg in Verilog? *1/1

- ○ bit
- ◉ logic ✓
- ○ wire
- ○ int

✓ What is the key difference between a function and a task in SystemVerilog? *1/1

- ○ A function can have delays, but a task cannot
- ◉ A task can have delays, but a function cannot ✓
- ○ A function can call a task, but a task cannot call a function
- ○ Functions return multiple values, while tasks return a single value

✓ Which of the following statements about SystemVerilog tasks is true? *1/1

- ○ They cannot include delays
- ○ They cannot call functions
- ○ They can return a value
- ◉ They can call other tasks ✓

✓ Which of the following statements is true about SystemVerilog functions? *1/1

- ○ They can contain delays
- ○ They cannot return a value
- ◉ They must execute within a single time step ✓
- ○ They can contain blocking assignments only

✓ What is the default return type of a SystemVerilog function? * 1/1

- ○ int
- ○ bit
- ○ void
- ◉ logic ✓

✓ Which of the following is NOT a type of code coverage? * 1/1

- ○ Statement coverage
- ○ Branch coverage
- ○ Functional coverage
- ◉ Power coverage ✓

✓ What is the purpose of statement coverage? * 1/1

- ○ Ensure all possible paths in a design are tested
- ◉ Verify that every executable statement in the code is executed at least once ✓
- ○ Check that each signal transition is exercised
- ○ Reduce simulation time

✓ Which of the following coverage types ensures all conditional branches are executed? *1/1

- ○ Statement coverage
- ◉ Branch coverage ✓
- ○ Toggle coverage
- ○ Path coverage

✓ Which of the following is a major driver for the increased importance of digital verification? *1/1

- ○ Increasing complexity of IC designs
- ○ Decreasing time-to-market pressures
- ○ Rising cost of design errors
- ◉ All of the above ✓

✓ Why is early verification in the design cycle important? * 1/1

- ○ To reduce the cost of fixing bugs later
- ○ To improve design quality
- ○ To meet tight deadlines
- ◉ All of the above ✓

✓ Which datatype is suitable for modeling signals with unknown or high-impedance states? *1/1

- ○ bit
- ○ int
- ◉ logic ✓

✕ What is the default datatype in SystemVerilog if no explicit datatype is specified? *0/1

- ○ bit
- ○ logic
- ◉ int ✕
- ○ reg

Correct answer

- ◉ logic

✓ How is a non-void function called in SystemVerilog? * 1/1

- ○ As a standalone statement.
- ◉ As an expression within an assignment. ✓
- ○ Using the fork join statement.
- ○ Using the begin end statement.

✓ How is a task called in SystemVerilog? * 1/1

- ○ As an expression within an assignment.
- ◉ As a standalone statement. ✓
- ○ Using the generate statement.
- ○ Using the package statement.

✓ What is the purpose of using subroutines in SystemVerilog? * 1/1

- ○ To improve code reusability
- ○ To simplify complex logic
- ○ To enhance code readability
- ◉ All of the above ✓